

Editorial

Information systems development (ISD), at the core of the information systems discipline, is an evolving field, faced with persistent challenges due to rapidly changing social and business environments as well as emerging technologies and technical infrastructures. Many of these issues have been discussed in the *Information Systems Journal* (see, for example, Kautz *et al.*, 2007).

Developments such as the Web 2.0 based on interactivity and graphical multimedia features, mobile application development for handheld devices, business applications based on standardised software architectures and software-as-services offered as part of a cloud pose new challenges to the development of IS products. Global and distributed organisations, agile software development and the reversal of outsourced and offshored development (insourcing) impact how information systems (IS) are designed, developed and delivered. These trends lead to new or adapted IS products, which have to be developed under new conditions, and these changes add complexity to an already complex problem space wherein, despite 50 years of ISD experience, the perception of the so-called 'software crisis' still persists. Unfinished and runaway projects, information systems poorly aligned with businesses and user requirements, and the resources required to develop ISD are still major concerns.

Research in the field is largely fragmented, and practice often seems to be ahead of research. Conversely, where research is indeed ahead, industrial uptake of academic research results is often limited. As a consequence, there is an urgent need for an extant and integrative theory based on extensive, empirical field research. The aim of this special issue is to publish new research, which investigates the new trends in ISD from both a product and a process perspective and which goes beyond surface-level considerations.

There is a general paucity of theory-building in ISD research; theory and studies of longitudinal processes of organisation, specialisation and institutionalisation in ISD are needed. Little ISD research goes beyond ISD methods; there is a need for theory and studies about social behaviour and processes of communication, negotiation, and learning and the relation to the broader historical, political and social context of ISD. Finally, there is a paucity of ISD research that relates individual knowledge, learning and sense-making to the broader context; this kind of theory and related studies are also needed. We have therefore called for research addressing questions such as: What are the different types of ISD relating to social, commercial, organisational and technological contexts? How is this diversity dealt

with? What are the different kinds of knowledge and skills needed in different types of ISD contexts? What new organisational structures have evolved to address the new ISD? What are the individual mechanisms for dealing with the diversity of environments and applications? How does learning and sense-making take place in the new social, commercial, organisational and technological contexts? With this special issue we hope to motivate rigorous research examining the effectiveness of approaches to ISD, including associated managerial aspects, using sound theoretical frameworks and appropriate research methods.

We see the practice of ISD as including activities such as analysis, encompassing problem formulation and requirements elicitation, design, programming and maintenance. We also include in the realm of ISD the organising and management of ISD projects and the creation and use of ISD methods and tools for the development, deployment and organisational and/or societal implementation and utilisation of information technology. This expansive view of ISD covers a multitude of approaches. In addition it accepts notions such as amethodical open-source software development, extreme programming, service-oriented architecture, component-based development, offshoring, commercial off-the-shelf software, cloud computing (including private or enterprise cloud development) and web services, mobile application development and development for Web 2.0 as ISD practices influencing the wider discourse.

As a reaction to our call for contributions we received numerous submissions and based on a rigorous reviewing process selected four of them for inclusion in this issue. The papers address a variety of ISD issues: developing user requirements, characteristics of control in different approaches to ISD, mindfulness and prerequisites for agility, and boundary resource design and use in third-party development.

In the first paper, 'The Emergence of Shared Understanding: Applying Functional Pragmatics to Study the Requirements Development Process', Corvera Charaf, Rosenkranz and Holten report from their study of communication in an ISD project based on an observational field study. Departing from the conceptualisation of ISD as an interactive development and formalisation process, they focus on linguistic communication and the development of language-based shared understanding of requirements. They adopt functional pragmatics and examine how linguistic communication is shaped and regulated during requirements development in the project and analyse the semantic alignment process by which stakeholders achieve shared understanding in requirements development. By developing a pattern for semantic alignment and its use as an analytical lens, they provide a systematic and insightful understanding of communication in information systems development and present their results in the form of eight propositions.

The second contribution, 'Controlling Information Systems Development: A New Typology for an Evolving Field' by Cram and Brohman, explores how and why ISD approaches differ in their tactics to control the behaviour of development teams. Drawing from research on organisational control, they conducted qualitative case studies at four organisations to examine the extent that traditional and contemporary ISD approaches can highlight a unique combination of control attributes. The applied development approaches ranged from the waterfall systems

development approach to a hybrid systems development methodology to an agile development approach. Based on the data collected, they build an ISD control typology that differentiates ISD approaches on the basis of control objectives, product- or process-focused, and control practices, preventive- or detective/corrective-focused. Product-related control objectives and preventive control practices seem to fit with an ISD approach that is highly structured, such as waterfall, while process-focused objectives and detective and corrective practices seem to fit with more flexible development approaches, such as agile and hybrid development approaches and may be most effectively used where product objectives and detective/corrective practices or process objectives and preventive practices are combined. The proposed typology can be used by practitioners to inform and guide more effective ISD control choices, while providing researchers with a new model that recognises the unique control aspects of today's ISD approaches.

The next article by McAvoy, Nagle and Sammon, 'Using Mindfulness to Examine ISD Agility', proposes that the presence of the prerequisites for ISD agility should be identified in software development team members, through examining their actions and behaviours, as opposed to only examining the ISD method components, the agile practices themselves. In the article, based on an ongoing longitudinal case study of an ISD organisation and its ISD process, they argue that this departure can be facilitated through the promotion of organisational mindfulness, where, at a conceptual level mindfulness requires a continuous attention to detail and vigilance to minimise errors and respond effectively to unexpected events. Mindfulness strives to promote the prerequisites for ISD agility, where agile practices themselves may fall short. The examination of software development teams' behaviours over methods shifts the focus away from particular ISD methods and associated practices and towards ISD team capabilities, mindful behaviours, and provides an insight into the practice of what the authors term 'doing' Agile and 'being' Agile and demonstrates the value of the concept of mindfulness for ISD agility.

The final paper, 'Balancing Platform Control and External Contribution in Third-Party Development: The Boundary Resources Model' by Ghazawneh and Henfridsson, takes an organisational perspective on ISD and documents the significance of using platform boundary resources such as application programming interfaces to cultivate platform ecosystems for a new development approach, namely third-party application development. For this context, the authors propose a theoretical model that centres on two drivers behind boundary resources design and use – resourcing and securing – and how these drivers interact in third-party development. The model is applied to a detailed case study of Apple's iPhone platform. The application of the model serves as an illustration of the model's plausibility and generates insights about the conflicting goals of third-party development: the maintenance of platform control and the transfer of design capability to third-party developers. The authors generate four specialised constructs for understanding the actions taken by stakeholders in third-party development: self-resourcing, regulation-based securing, diversity-resourcing and sovereignty securing. This research extends and complements existing platform literature and contributes new knowledge about an alternative form of ISD.

We believe that the selected articles demonstrate the width and depth as well as the evolution of ISD as a research field, and we are confident that they contribute to new insights with regard to new trends in ISD.

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REFERENCE

- Kautz, K., Madsen, S. & Nørbjerg, J. (2007) Persistent problems and practices in information systems development. *Information Systems Journal*, **17**, 217–239. doi: 10.1111/j.1365-2575.2007.00222.x.

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